Examination Report
Physical Science
6888

EGCSE

2025

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Paper 6888/01

General Comments

The total number of candidates who sat for this paper was about 12341.

This shows a slight increase from the previous year's exam. This year's performance was poor compared to last year. Most candidates obtained marks less than half of the total marks for the paper. Very few candidates were above 30. There was an observable decrease in the number of candidates who got zero even though they had attempted most of the questions. This shows a significant drop from the number of zeros recorded in the previous year.

Questions that were more challenging to most candidates were Questions **5(b), 7, 8(a), 10(a)** and **14**. Question **14(b)** proved to be the most challenging because very few candidates managed to get full marks. Questions that were accessible to most candidates were **1(a), 4(a), 6,** and **12(b).**

Comments on Specific Questions

Question 1.

(a) This question was one of the most accessible questions to candidates. Candidates were asked to name the change of state that occurs when iodine gas changes directly to a solid. Common wrong responses include disposition, condensation, decomposition and common wrong spelling include sublimentation, sublimantion were not awarded a mark.

Expected correct response: *deposition/ sublimation*

(b) This question was challenging to most candidates. Candidates were asked to draw the arrangement of particles in solid iodine in Fig. 1.1. Common wrong responses: different shapes and sizes of particles, some draw arrangement for gases and liquids

Expected correct response: *particles of same size and same shape, regular arrangement with all touching and touching the base of the container*

Question 2

(a) This part of the question was accessible to most candidates. Candidates were given a diagram showing a pendulum with a bob swinging between points **A** and **C**. Candidates were asked to define the *period* of the pendulum, referring to

Fig. 2.1. Common wrong responses: movement of the bob, distance moved by the bob, per unit time, others just defined a period without referring to Fig. 2.1

Expected correct response: *time taken by the bob to move from **A** to **C** and back to **A**.*

(b) This question was moderate. Candidates were asked to calculate the frequency of the pendulum that makes 20 complete swings in 15 seconds and state the unit. Common wrong responses: wrong unit, wrong formula, wrong substitution

Expected correct response: *frequency =* $\frac{\textit{number of swings}}{\textit{time}}$

=1.3 Hz or 1.3 Hertz

Question 3

(a) This question was the most challenging to most candidates. Candidates were asked to explain why C*l*–35 and C*l*–37, isotopes of chlorine, have an overall charge of zero. Common wrong responses: have same number of positive and negative charges, same protons and electrons and some candidates just defined isotope.

Expected correct response: *same number of protons and electrons*

(b) This question was challenging to most candidates. Candidates were asked to explain why C*l*–35 and C*l*–37, isotopes of chlorine, have the same chemical properties. Common wrong responses: have same number of electrons, are in the same group, have the same number of shells, are halogens, are the same elements.

Expected correct response: *same number of electrons in the outer shell*

Question 4

(a) This question was accessible to most candidates. Candidates were given iron and nickel as examples of magnetic materials. Candidates were asked to state what is meant by magnetic materials. Common wrong responses: easily magnetized and demagnetized, can attract and repel other metals, can attract and repel magnetic materials

Expected correct response: *a substance that can be attracted by a magnet*

(b) This part was also challenging for most candidates. Candidates were given a diagram showing two magnets with North poles facing each other. They were required to draw magnetic field lines between the two poles. Common wrong responses: dotted lines, lines crossing, wrong direction.

Expected correct response: *solid lines showing repulsion, North to South Direction*

Question 5

Candidates were told that the Group VII elements of the Periodic Table are called halogens.

(a) This part of the question was challenging to most candidates. Candidates were required to state the colour and state of matter of bromine at room temperature and pressure. Common wrong responses: for the colour they stated reddish brown, dark brown, yellow brown, brick red and for the state it was solid

Expected correct response: red-brown, liquid

(b) This part was also generally challenging to most. Candidates were asked to suggest the colour change observed when bromine is added to aqueous potassium astatide. Common wrong responses: most candidates stated a colour instead of colour change. e.g. brown, blue black, brick red.

Expected correct response: *from red brown to black*

Question 6

This question was generally well answered since a very high number of candidates managed to get both marks. Candidates were required to calculate the work done on a trolley of mass 1.2 kg that is moved by a 1.5N force over a horizontal floor of distance 4m. Common wrong responses: W = f x d, W = F x D, Wd = mgh, F = ma, some candidates used wrong substitution.

Expected correct response: *W = F x d / W = 15 x 4,*
= 60

Question 7

This question was challenging as very few candidates managed to get at least 1 mark. Most candidates confused the concept of preparation of insoluble salts with that of soluble salt. Candidates were required to describe how pure dry crystals of manganese hydroxide, a white precipitate, can be obtained from reacting manganese nitrate and manganese hydroxide solutions. Common wrong responses: use simple distillation, crystallization, dry under the sun, some candidates were listing the processes without description i.e. filter, wash, dry.

Expected correct response*: filter to obtain crystals of manganese hydroxide, rinse with distilled water to remove impurities, dry crystals between filter papers/ in an oven*

Question 8

Candidates were told that the lower and upper fixed points of a liquid-in-glass thermometer are determined during calibration.

(a) This part of the question was challenging. Candidates were asked to describe the lower fixed point of a thermometer. Common wrong responses: lowest reading on a thermometer, 0 ˚C, temperature of melting ice, temperature of freezing water.

***Expected correct response**: temperature of pure melting ice taken as 0 ˚C*

(b) This part of the question was accessible. Candidates were asked to state the effect on sensitivity of using a wider capillary tube in a liquid-in-glass thermometer. Common wrong responses: become slower, sensitivity is low, sensible, increases

***Expected correct response**: sensitivity decreases*

Question 9

(a) This question was challenging to most candidates. Candidates were given a table showing observations made and products obtained, at the cathode during the electrolysis of concentrated aqueous sodium chloride and aqueous copper(II) sulfate. They were required to fill the empty boxes in Table 9.1 by writing the product for electrolysis of concentrated aqueous sodium chloride and the observation for electrolysis of aqueous copper(II) sulfate. Common wrong responses: chlorine, sodium for product, and bubbles, precipitate for observation

***Expected correct response**: hydrogen gas as product for concentrated sodium chloride, and brown layer as observation for copper(II)sulfate*

(b) This part was also challenging for most candidates. Candidates were asked to suggest the pH value of the solution formed at the end of the electrolysis of concentrated aqueous sodium chloride. Common wrong responses: alkaline, pH 7, pH 9

***Expected correct answer**: any value from 12 to 14*

Question 10

(a) This part of the question was one of the most challenging to most candidates. Candidates were given a diagram showing wavefronts for water waves travelling from shallow to deep water. They were asked to define the term wavefront. Common wrong responses: distance between crests/ waves

***Expected correct answer**: a line joining all points that are in phase in waves.*

(b) This part of the question was challenging to most candidates. Candidates were asked to draw three wavefronts on Fig. 10.1 showing the waves as they move from shallow to deep water. Common wrong responses: no bending, bending backwards, sine wave, unequal wavelength in deep water, diffraction.

Expected correct response: *three parallel waves refracted/ bending forward, waves with larger and equal wavelength in deep water*

Question 11

This question was generally challenging to most candidates. Candidates were asked to explain why ethanol, a covalent compound, is more volatile than calcium chloride, an ionic compound. Common wrong responses: weak bonds in ethanol, some did not compare the energy but only stated that a lot of energy is required, some referred to particles instead of molecules and ions.

Expected correct response*: weak intermolecular forces in ethanol, less energy required to break the forces/ strong electrostatic forces between ions in calcium chloride, more energy required to break the forces*

Question 12

Candidates were told that α-particles, β-particles and γ-rays are three types of radioactive emissions.

(a) This part of the question was moderate. Candidates were asked to describe the relative penetrating ability of the β-particles. Common wrong responses: β-particles pass through lead, through thick paper, cardboard. Most candidates just compared the penetrating ability of α-particles, β-particles and γ-rays but did not mention paper and aluminium.

Expected correct response*: pass through paper but are stopped by aluminium foil*

(b) This part of the question was accessible to a number of candidates. Candidates were required to state one safe way of handling radioactive materials. Common wrong responses: wear protective clothing, wear googles, wear sunglasses.

Expected correct response: *use tweezers to handle, store in lead containers, wear lead apron*

Question 13

Candidates were given a diagram showing a simple cell made of copper and zinc metal electrodes dipping into dilute sulfuric acid.

(a) This question was moderately answered. Candidates were asked to draw an arrow on the diagram to show the direction of electron flow. Common wrong responses: arrow in opposite direction, electrons flowing through acid

Expected correct response*: electrons moving from zinc to copper through external circuit*

(b) This part was also fairly done. Candidates were asked to state a change that can be made to the cell to increase the brightness of the bulb. Common wrong responses: add more cells, increase current, add more bulbs, increase temperature, replace electrodes with carbon or platinum electrodes.

Expected correct response*: replace zinc with magnesium/ replace zinc with a more reactive metal*

Question 14

(a) This part of the question was accessible to most candidates. Candidates were asked to state the type of material used to make LEDs. Common wrong responses: tungsten, diode, plastic

Expected correct response: semiconductor material

(b) This part of the question was challenging to most candidates. They were required to explain how LEDs produce light. Common wrong responses: holes fuse with electrons, some described production of light in fluorescence and solar. Some candidates did not mention the loss of energy by electrons.

Expected correct response: *electron collides with a hole and loses energy in the form of a photon*

Question 15

Candidates were given the equation for the reaction of methane with lead(II) oxide to produce lead, carbon dioxide and water.

(a) This part of the question was accessible to a number of candidates. Candidates were asked to explain in terms of oxygen transfer, why lead(II) oxide is reduced. Common wrong responses: lead(II) loses oxygen, lead loses electrons, lead(II) oxide loses oxygen, losses oxygen molecules.

Expected correct response: *loses oxygen*

(b) This question was moderate. Candidates were asked to state how carbon dioxide gas causes global warming. Common wrong responses: destroys the ozone, absorbs UV / light / sun's radiation / rays, forms carbon monoxide.

Expected correct response: *absorbs more heat than it emits back to space*

Paper 6888/02

Structured Theory Paper

General Comments

12 564 candidates registered for this component but the number of candidates who actually sat for the component was 12 331. These numbers showed an increase of approximately 1 400 candidates who actually sat for the component.

The paper was marked out of a total of 80 marks. The highest attained score was 72 marks which is 7 marks higher than the highest score of 65 marks attained in the previous year. Compared with the previous year, the performance was better in terms of the highest mark scored. There was quite a noticeable number of candidates who scored more than 30 marks compared to the previous year. However, there was still a noticeable of candidates who scored single digit marks. The majority of the candidates scored between 15 – 30 marks, which showed a higher range and better performance than the 15 – 25 marks which was scored by the majority of candidates in the previous year.

Most candidates continue to have challenges with questions that require application, description and explanation. Candidates displayed confidence with recall questions and calculations, however, most candidates lost marks for failure to round off to the number of significant numbers required by the syllabus. Candidates should avoid rounding off unnecessarily. If the answer is not exact, rounding off should be done to 3 significant figures as stated in the syllabus not unless the accuracy is stated in the question. There was a better performance on calculations on stoichiometry compared to the previous years.

Questions that proved particularly easy for most candidates were: **1, 2 (a) (i), 2 (a) (iii), 3 (a), 3 (d), 4 (b) (i), 4 (b) (ii), 5 (a), 8 (b) (i), 11 (a), and 11 (c) (i)**. Questions that proved particularly difficult for most candidates were: **4 (c), 5 (b) (ii), 5 (c), 7 (b), 8 (a), 9 (c), 10 (b) (ii), 11 (c) (ii), and 11 (d)**.

Comments on Specific Questions

Question 1

This question was generally well done by most candidates. Most candidates were able to score at least 2 marks out of the available 4 marks. Candidates were given five structures, **A**, **B**, **C**, **D**, and **E** and were required to give a structure that corresponds to a given property. There was no pattern of common wrong responses.

Expected responses:

(a) ***C***

(b) ***A***

(c) ***D***

(d) ***E***

Question 2

(a) **(i)** This was a fairly done question as most candidates were able to obtain at 1 mark out of the maximum 2 marks. Candidates were required to state what is meant by a *vector quantity*. A common wrong response was that of defining the term *velocity*.

Expected response: *a measurable quantity with magnitude and direction*

(ii) This was another fairly done question. Most candidates were able to state, partly, why the girl was travelling at a constant speed. Most candidates did not fully answer the question as a result most candidates obtained one mark out of two from this question. Common wrong responses included: weight is equal to the air resistance, the forces are the same, the road was flat and the girl was travelling at terminal velocity.

Expected response: *forward driving force is equal to the total backward force*
making the resultant force to be zero

(iii) This part question was also fairly done. Average and high scoring candidates were able to score both marks. The majority of the candidates were able to obtain at one mark out of the two available marks. Candidates were required to name the forces that cause the total backward force. Common wrong responses were: gravitational force, weight, thrust, resultant force and resistance force.

Expected response: *air resistance and friction*

(b) **(i)** This part question was challenging for most candidates as the two marks was scarce. Candidates were required to describe the motion of the girl, from the given velocity-time graph, during the first 400 seconds. Candidates struggled to read the scale of the graph, and this led to loss of marks. The common wrong response was constant acceleration from 0 seconds to 200 seconds and terminal velocity from 200 seconds to 400 seconds.

Expected response: *constant acceleration from 0 s to 250 s*
constant velocity/ speed from 250 s to 400 s.

(ii) Generally, this question was challenging for most candidates, however most high scoring candidates were able to obtain all the marks. Candidates were required to determine the distance covered by the girl from 400 seconds to 500 seconds. A common wrong response was that of using the formula: $\text{speed} = \frac{\text{distance}}{\text{time}}$. Some candidates determined the distance covered for the whole journey instead of the last 100 seconds and they lost the marks.

Expected response: *distance = area under graph / ½ bh / ½ x 100 x 8*
= 400 m

Question 3

(a) This was a well-done question as most candidates were able to score this mark. Candidates were given a chromatogram of spinach juice, and they were required to name the most soluble dye. ***The expected response*** of *carotene* was very common. Common wrong responses were acetone and chlorophyll-b.

(b) This was a fairly done question as most of the candidates were able to score at least one mark from the available two marks. Candidates were required to calculate the retardation factor given the formula $R_f = \frac{\text{distance moved by solute}}{\text{distance moved by solvent}}$

Most candidates lost a mark for failure to round off to 3 significant figures as the answer was not exact. Candidates also lost marks for writing units.

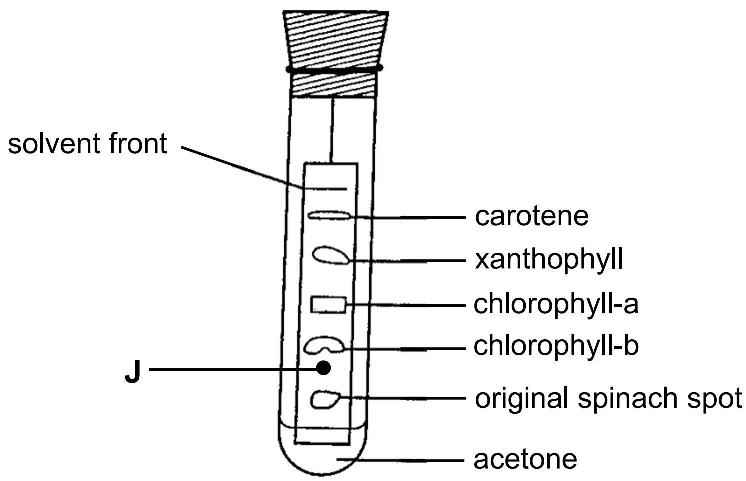
Expected response: $R_f = \frac{8}{12}$
0.667

(c) **(i)** This part question was challenging for most candidates as most candidates failed to score the mark. Candidates were required to describe how to identify, on the chromatogram, a component of spinach that is colourless. Even candidates who had the idea of a locating agent failed to describe how to use it. A common wrong response was: use a locating agent. Candidates who wrote: use locating agent or use UV radiation without stating how to use it did not earn the mark.

Expected response: *spray chromatogram with a locating agent/ expose chromatogram to UV radiation*.

(ii) This was another poorly done question as most candidates failed to mark the position of lutein, which is least soluble, on the chromatogram. A noticeable number of candidates left this question unattempt. Common wrong responses involved marking the position of lutein below the original spinach spot or above carotene.

Expected response: J *marked between chlorophyll-b and the original spot.*



(d) This part question was well done as most candidates were able to explain that spinach juice is not a pure substance because *it forms multiple spots on the chromatogram*. Common wrong responses were that it contains two or more elements and that it has more than one colour.

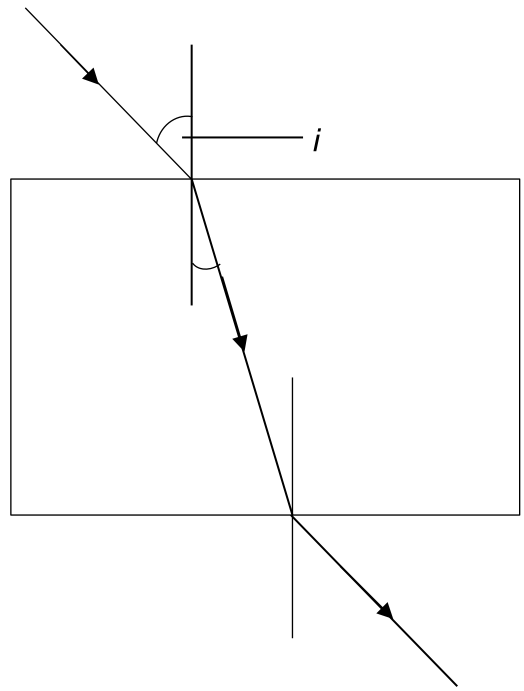
Question 4

The whole of this question was generally well done. The question was on refraction and the electromagnetic spectrum.

(a) **(i)** This question was well done. The majority of the candidates were able to mark the angle of incidence between the normal and the incident ray on the given figure. A common wrong response was that of marking more than one angles.

(ii) This part of the question was fairly done. A noticeable number of the candidates were able to score at least one mark out of the available two marks. Candidates were required to complete the path of the incident ray as it enters and leaves the glass block. Common wrong responses were not putting arrows to indicate direction and the emergent ray not being parallel to the incident ray.

Expected response: *refracted ray bending towards the normal*
emergent ray parallel to the incident ray



(b) (i) This was another well-done question as the majority of the candidates were able to correctly identify regions **K** and **L** of the electromagnetic spectrum as *infra-red* and *X-rays,* respectively. Common wrong responses were infra-red and ex-rays.

(ii) This part question was also well-done. The majority of the candidates were able to score this mark. Candidates were required to name the member of the electromagnetic spectrum with the shortest wavelength. Common wrong responses were radio waves and microwaves.

Expected response: *gamma rays*

(iii) This part question was also well-done. Most candidates were able to score the mark. Candidates were required to state one property common to all electromagnetic waves. Candidates lost the mark for failure to write complete properties e.g., they all obey the wave equation instead of; they all obey the wave equation, $v = f \times \lambda$. A common wrong response was that of giving the wave equation as $c = f \times \lambda$.

Expected response: *transverse in nature/ they all obey the wave equation:*
$v = f \times \lambda$ / they can be reflected, refracted and diffracted/
all travel with the same speed, 3×10^8 m / s, in vacuum.

(c) This part question was challenging for most candidates as very few candidates were able to score both marks. Candidates were required to explain why sound cannot be transmitted in a vacuum. Common responses that did not earn a mark were; there are no air particles, there is an empty space, sound vibrates and sound travels through compressions and rarefactions.

The expected response: *there are no particles in a vacuum to vibrate*

Question 5

This question was about the manufacture of ammonia in the Haber process. Generally, the whole question was poorly done.

(a) This part question was well-done as the majority of the candidates were able to recall the process used in the manufacture of ammonia as the *Haber* process. A common wrong response was hyber process.

(b) **(i)** This part question was poorly done as the majority of the candidates failed to score even a single mark from the available two marks. Candidates were required to name the catalyst used in the manufacture of ammonia. Common wrong responses were phosphoric acid, aluminium oxide and nickel.

Expected response: *iron/ Fe*

(ii) This was another poorly done question. Candidates were required to explain, using the collision theory, how the catalyst increases the rate of the reaction. The majority of the candidates failed to show understanding of the collision theory. There were no common wrong responses.

Expected response: *the catalyst reduces energy needed for successful collisions /*
provides an alternative pathway/route with a lower activation energy more
collisions result in a reaction / more successful collisions <u>*per unit time*</u>

(c) This was another very challenging question for the majority of the candidates. Even most high scoring candidates were able to score only one mark out of the maximum three marks. Candidates were required to explain the effect of increasing the temperature to 600 °C on the yield of the ammonia. The majority of the candidates confused the question with the kinetic particle theory. Most candidates made references to enzymes being denatured.

Expected response: *yield is reduced, reaction to form products is exothermic*
equilibrium shifts left / favours the backward reaction to oppose the
increase in temperature

Question 6

This question was on electromagnetic induction. Generally, the question was challenging for most candidates.

(a) This was a poorly done question yet it was a simple recall question. The majority of the candidates failed to name the principle demonstrated by the moving conductor between the poles of a magnet. Common wrong responses were: magnetic induction, electromagnet and Fleming's rule.

The expected response: *electromagnetic induction*

(b) This was another challenging part question for most candidates. Candidates were required to state and explain the effect of moving the conductor vertically upwards between the poles of the magnet. Most candidates failed to score the marks because of failure to express themselves which led to wrong science e.g., galvanometer moves instead of galvanometer shows a deflection, current is produced instead of current is induced. Another common wrong response was that unlike poles attract.

Expected response: *galvanometer shows a deflection*
conductor cuts magnetic field (causing a change in magnetic flux)
an emf or current is induced (in the circuit)

(c) This question was also challenging for most candidates as most candidates failed to score the mark. Candidates were required to state the effect, on the galvanometer, of moving the conductor at a higher speed. A common wrong response was that the galvanometer shows a faster deflection.

Expected response: *larger deflection / larger current / more emf induced*

(d) This question was fairly done. Most candidates were able to score at least one mark out of the available 2 marks. Candidates were required to state and explain the effect, on the galvanometer, of leaving the conductor stationary between the poles of the magnet. A common wrong response was that the galvanometer shows no reading.

Expected response: *no deflection*
conductor does not cut magnetic field lines

(e) This part question was well done. The majority of the candidates were able to score the mark. Candidates were required to name a device that uses the principle of electromagnetic induction. Common wrong responses were: a.c. motor, ammeter and hair drier.

Expected response: *(ac/dc) generator / dynamo / alternator*

Question 7

(a) This was another challenging question for most candidates as the correct response was not common. Candidates were required to write the electron configuration of a lithium ion. A common error was to ignore the ion, and candidates were giving the electron configuration of lithium atom. Common wrong responses were: 2.1/ 2.5 and ${}^{7}_{3}Li$

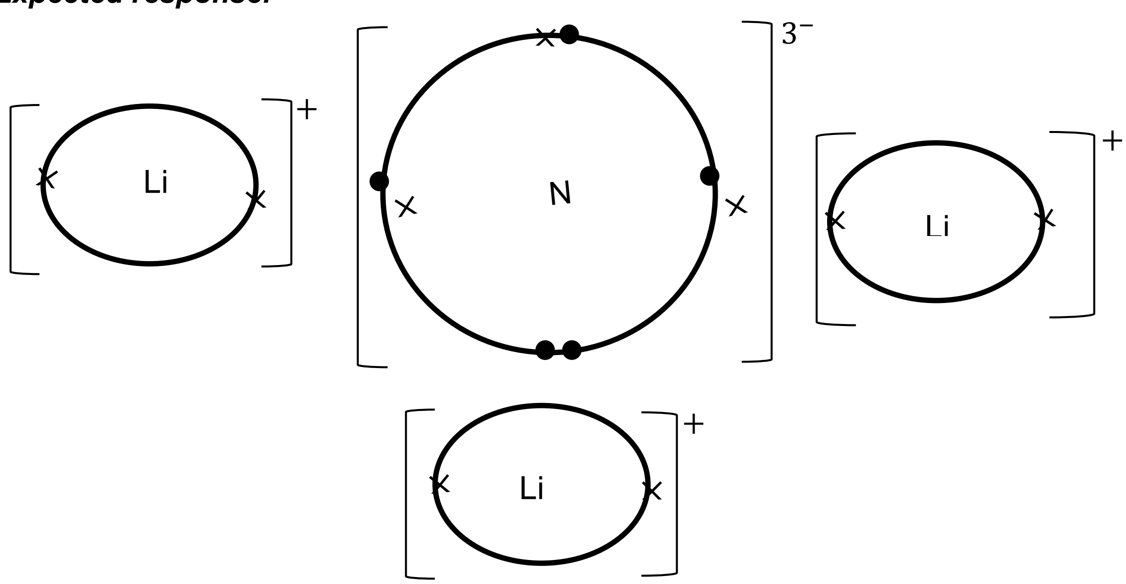
***Expected response**: 2 OR 2.0*

(b) This was a poorly done question as only one mark was accessible to the majority of the candidates. Candidates were required to explain why lithium atoms are very reactive. Common wrong responses included: lithium has fewer electron shells and lithium has few electrons in the outer shell than nitrogen.

***Expected response:** one electron on outer(most) shell that is weakly attracted to nucleus / easy to lose*

(c) This question also proved challenging for most candidates. Only a minority of the high scoring candidates got all the marks. Candidates were required to draw a "dot and cross" diagram to show the bonding in Li_3N. The majority of the candidates attempted covalent bonding instead of ionic bonding.

Expected response:



(d) **(i)** This part question was fairly done. The majority of the candidates were able to score at least two marks out of the available three marks. Candidates were required to calculate the empirical formula of the oxide of nitrogen, N_xO_y. Common wrong responses were: N_2O and NO.

Expected response: *nitrogen : oxygen*

$$1.4 : 3.2$$

$$\frac{1.4}{14} : \frac{3.2}{16}$$

$$0.1 : 0.2$$

$$\frac{0.1}{0.1} : \frac{0.2}{0.1}$$

$$1 : 2$$

empirical formula is: NO_2

(ii) This was a challenging part question for most candidates as the two marks were scarce for most candidates. Candidates were required to determine the formula of an oxide of nitrogen with a molecular mass of 92. There were no common wrong responses.

Expected response: *empirical mass = 46*

$$\frac{92}{46} = 2$$

molecular formula = N_2O_4

Question 8

This question tested candidates' comprehension of electricity.

(a) This question was poorly done as both marks were scarce to score by the majority of candidates. Candidates were required to describe how heat is produced in the element of an electric heater. The concept of high resistant element was not common with many candidates. Common wrong responses were: chemical energy is converted to electrical energy and then to heat energy, light energy is converted to heat energy. Mentioning of a bimetallic strip was also very common.

Expected response: *current flow experiences resistance (in the element)*

electrical energy converted to heat energy

(b) **(i)** This was a well done question. The majority of the candidates were able to calculate the current in the heater when working normally. Use of $I = \frac{V}{R}$ was common and was not acceptable. Candidates also lost marks for use of $I = \frac{W}{V}$ where they used *W* for power.

Expected response: $I = P / V = 1200 / 240$

$= 5\ A$

(ii) This was a fairly done question. Most candidates were able to score at least a mark. Candidates were required to describe how a fuse protects the appliance and the user. Candidates lost marks for failure to express themselves or to use correct scientific terms, for an example: fuse reduces current, fuse burns instead of fuse melts and excessive voltage instead of excessive current.

Expected response: *fuse melts when abnormally high current flows and breaks the circuit*

(c) This was another poorly done question. Most candidates failed to score this mark.
Candidates were required to state one hazard of broken fluorescent lamps. Mentioning of mercury was scarce. A common wrong response was that fluorescent lamps contain dangerous and harmful gases. Another common wrong response was that mercury causes cancer.

Expected response: *(fluorescent lamps) contain mercury which is poisonous*

Question 9

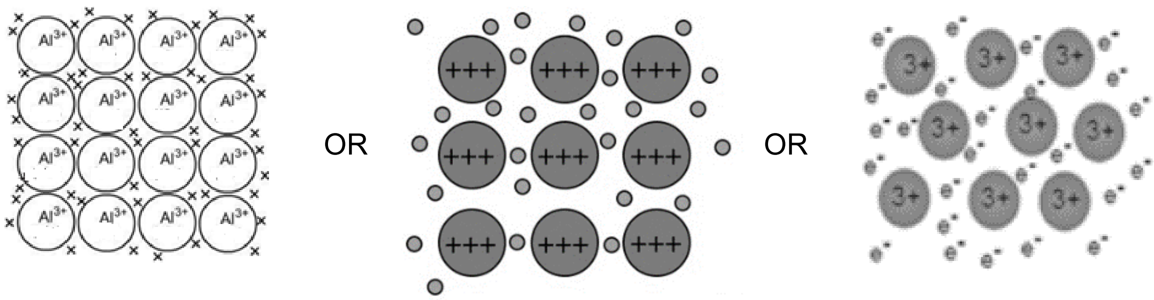
(a) This question was well-done as most candidates were able to state the term used to describe oxides that behave both as an acid and a base as *amphoteric.* Spelling mistakes were common and some lead to loss of the mark.

(b) This was a challenging question for the majority of the candidates as most of them loss this mark. Candidates were required to name a reagent that reacts with aluminium oxide when it behaves as a base. Some candidates had an idea that it is an acid but did not name any acid.

Expected response: *any named acid*

(c) This was a challenging question for most candidates. Two marks out of the available three marks was scarce. Candidates were required to draw a diagram to show metallic bonding in aluminium. The 3^+ charge for aluminium was scarce in the candidates' responses. Representation of delocalized electrons was also scarce in candidates' responses.

Expected response: *circles (atoms) in rows*
aluminium ions with a charge of 3^+
delocalised electrons

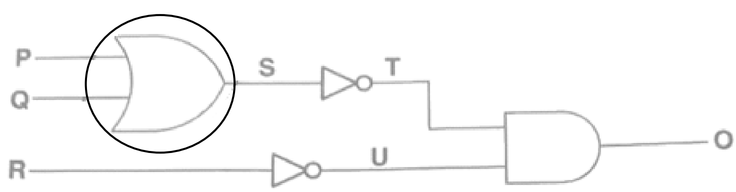


Question 10

Generally, the whole of this question was poorly done. The question assessed understanding of logic gates.

(a) This was a fairly done question as a good number of the candidates was able to score the mark. Candidates were required to circle an **OR** logic gate from a figure that showed a combination of logic gates. A common wrong response was a **NOT** gate. Candidates also lost the mark for circling more than one logic gate. Some candidates circled the letters **P** and **Q** representing the inputs not the logic gate itself, and they lost the mark.

Expected response:



(b) **(i)** This question was challenging for most candidates as only a minority got the three marks. Even one or two marks was scarce. Candidates were required to complete a truth table for the given input combinations **P**, **Q** and **R**. A common wrong response was that of using digits like 2, 3 or 4 instead of binary digits 0 and 1.

Expected response:

P	Q	R	S	T	U	Output O
			1	*0*	*0*	*0*
			1	*0*	*1*	*0*
			0	*1*	*1*	*1*

(ii) This part question was also challenging for most candidates. The majority of the candidates did not realise that they had to use the truth table in **(b) (i)** to answer **(b) (ii)**.

Expected response: *0, 0, 0*

Question 11

The whole of this question was challenging for most candidates. The question was on organic chemistry.

(a) This part question was a simple recall and was well-done. The majority of the candidates were able to recall the process used to extract paraffin from petroleum as *fractional distillation*. Common wrong responses were: fractionating and distillation.

(b) This question was poorly done as most candidates failed to score even one mark.
Candidates were required to describe the conditions necessary to convert paraffin
into ethene. High temperature, high pressure and a catalyst were common wrong responses. Another common wrong response was: paraffin is flammable.

Expected response:

with respect to catalytic cracking:

pressure: 2-3 atmospheres /low pressure **accept** *catalyst: aluminium oxide*

temperature: 400 °C to 550 °C

OR

with respect to thermal cracking:

temperature: 700 °C to 900 °C

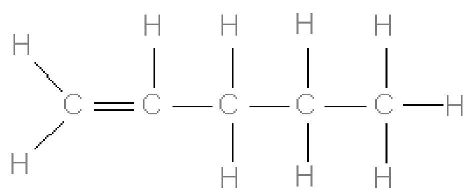
pressure: 30 atmospheres

(c) **(i)** This part of the question was well done as most candidates were able to name correctly the homologous series to which ethene belongs. Common wrong responses were: alkine, -ene and C = C.

Expected response: *alkene*

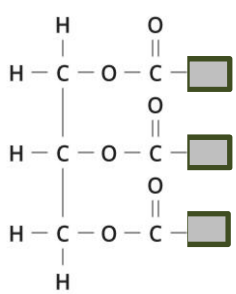
(ii) This was another challenging question for most candidates as the correct structure of pentene was not common in the candidates' responses. Candidates were required to draw the structure of the fourth member of the alkenes. A common wrong response was that of drawing the structure of butane and butene. Another common wrong response was that of drawing double bonds in all the carbons.

Expected response:



(d) This was one of the most challenging questions for candidates. Only a few candidates were able to score both marks. Candidates were required to draw the structure of the fat formed between glycerol and 3 fatty acids. A common error was that of drawing more than one functional group in the structure.

Expected response:



fat (triglyceride)

Paper 6888/03

Practical Test

General Comments

This Practical Test paper assesses the experimental skills and investigative skills of candidates which is Assessment objective C of the syllabus. The paper consists of two questions, one Chemistry and one Physics, each totalling 20 marks. The time allocated the paper seemed to be adequate as candidates attempted all questions. Candidates should be encouraged to carry out the experiments more than once if there is still time to spare as this assist candidates to make the correct observations and increases the reliability of results.

The number of candidates who sat for the paper increased slightly when compared to last year, 2024. The paper requires candidates to use scientific techniques, laboratory apparatus and materials including following a sequence of instructions. Candidates are also required to make and record observations and measurements, interpret and evaluate experimental observations. The skill of planning an investigation and evaluating methods used was also tested.

The performance of candidates was generally poor in describing observations. Terms like “few, moderate, many” were rare in describing the rates of reactions. Some candidates had difficulty recording observations in the correct answer space when given a table as a result interpreting their data became challenging. Some candidates showed a lack of familiarity with reading a thermometer.

In question 2, quiet a number of candidates could not convert the reading from centimetres to millimetres and converting mass to force even though the formula, W = mg was given. A number of candidates found it challenging to record in a table, they wrote their responses in the wrong space hence lost marks. Candidates’ graphing skills were generally poor when compared to previous years. The time allocated the paper seemed adequate since candidates were able to attempt all questions.

Comments on Specific Questions

Question 1

The question required candidates to investigate the reactivity of four metals and their reactions with dilute hydrochloric acid and aqueous silver nitrate by carrying out two experiments.

(a) In the first experiment candidates investigated the reactivity of the metal powders with dilute hydrochloric acid.

(i) Candidates were required to measure and record the temperature of dilute hydrochloric acid in the beaker initial temperature. This part of the question was generally well done as the majority of candidates could read the scale of the thermometer correctly. Some common wrong responses were 2.3, 2.5, 2.8 ˚C.

Expected response: supervisor's value ± 1 or 25± 1

(ii) This part of the question was generally well done. A number of candidates lost marks as hey failed to record their observations precisely, they did not show any comparison. Most common wrong response was bubbles produced in all test tubes. Other candidates lost two marks as their recorded temperature of copper was higher than that of magnesium.

***Expected responses**: temperature highest for magnesium, temperature lowest for copper, gas burns with a pop sound and all correct observations in the table*

experiment in test-tube	**final temperature of acid / ˚C**	**observation**
A		few bubbles/ slow reaction/red-brown solution/ green solution
B		moderate bubbles
C		no bubbles/ no change/ no reaction
D		many bubbles/ fast reaction

(iii) This part of the question was fairly well done. Candidates were asked to name the gas produced when magnesium powder reacted with dilute hydrochloric acid. Some wrong responses included ammonia, chlorine, carbon monoxide and carbon dioxide.

***Expected response**: hydrogen/ H_2*

(iv) This part of the question was fairly well done. Candidates had to identify the most reactive metal using the observations they made in **(a)(ii)**. Candidates who lost this mark either left the question unanswered or made a wrong choice even though their observations were correct.

***Expected response**: magnesium*

(v) This part of the question was fairly well done as candidates were able to justify their response for **(a)(iv)**. Candidates were asked to state two reasons for their response to **(a)(iv**). Most of the candidates who lost both marks did not give any comparisons in their response. They wrote incorrect responses such as fizz, temperature and bubbles.

***Expected response**: highest temperature change and most bubbles (more fizz / fastest reaction)*

(vi) This part of the question was fairly well done. Candidates were asked to identify a variable that was controlled to ensure the experiment was a fair test. Some common wrong responses included dilute acid, temperature, volume of metals, etc.

***Expected responses**: same mass of metals/ same concentration of acid / same initial temperature/ same volume of acid*

(b) In the second experiment candidates investigated the reactivity of the metals copper, magnesium, zinc and iron with aqueous silver nitrate.

(i) Candidates were expected to record their observations. This part of the question proved challenging to candidates as the majority of the candidates earned one mark out of the three marks. Some common wrong responses included temperature values being the same for all the reactions, reaction of magnesium with aqueous silver nitrate having the lowest temperature, colour of precipitate not given and the rate of the formation of the precipitate not referred to.

***Expected response**: temperature highest in magnesium and lowest in copper;*
iron – grey/ dark precipitate formed slowly
copper - grey/ dark precipitate formed slowly
magnesium - grey/ dark precipitate formed quickly

(ii) This question was challenging to candidates. Candidates were asked to state and explain the type of reaction that occurred in (b)(i). The most common wrong response is endothermic reaction. Quite a number of candidates were able to state the type of reaction but could not explain why.

***Expected response**: exothermic reaction because heat energy is released causing an increase in temperature.*

(iii) Candidates were informed that one of the products of zinc granules with aqueous silver nitrate is zinc nitrate. Candidates were asked to name the other substance formed. This question was poorly done as the majority of candidates gave wrong responses such as silver oxide, silver ions or silver hydroxide.

***Expected response**: silver metal*

(iv) This part of the question was fairly well done. Candidates were asked to identify the least reactive metal using their observations in **(b)(i)** and explain their answer. A number of candidates were able to state least reactive metal but could not explain why as a result they earned only one mark out of the two marks on offer.

***Expected response**: least reactive metal- copper*
explanation- grey solid formed slowest/ slowest reaction/ lowest maximum temperature

(c) This part of the question was fairly well done. Candidates were asked to use their results from (a) and (b) to arrange the four metals in order of their reactivity starting with the most reactive.

***Expected response:** most reactive – magnesium*
zinc
iron
least reactive - copper

Question 2

In this experiment candidates investigated the relationship between the load and the extension of a spring.

(a) Generally, the question was well done. Candidates were asked to read and record the position of the bottom of the spring**,** $\mathbf{X_o}$, in millimetres. The most common wrong response was 55 cm.

***Expected response**: supervisor's value ± 0.1 / 550 mm*

(b) This part of the question was fairly well done. Candidates suspended a 100 g mass on the spring. Then read and recorded the new position of the bottom of the spring, **X**, in millimetres. Some candidates recorded the value in a wrong position in the table which resulted in the loss of the mark.

***Expected response**: X greater than X_o (an increase in value)*

(c) Candidates repeated procedure **(b)** by adding 50 g masses, one at a time, on the suspended mass. Then recorded the new positions of the bottom of the spring for each 50 g mass added.

(i) This part of the question was poorly done. Candidates were asked to fill in the unit for extension. A number of candidates gave the units of moments, Nm, instead of units of length. Other wrong answers included kg, g, mg and even km.

***Expected response**: cm / mm*

(ii) This part of the question was poorly done. Candidates were asked to calculate the load, W, for each mass using the formula W = mg. A number of candidates multiplied the mass in grams instead of mass in kilograms resulting in wrong responses.

Expected response:

mass / kg	0.10	0.15	0.20	0.25	0.30
load / N	1.0	1.5	2.0	2.5	3.0

(iii) This part of the question was generally well done. Candidates were asked to calculate the extension, e, caused by each mass using the formula $e = X - X_0$. Some candidates lost marks for having negative extension.

***Expected response**: correctly calculated differences using candidate's values.*

(d) This part of the question was fairly well done. Candidates were asked to plot a graph of extension against load using the results in **(c)**. Some candidates had poor graphing skills, they used a scale that does not cover more than half the grid, points not clearly plotted and making a frequency polygon instead of drawing a line of best fit or smooth curve through the origin.

***Expected response**: correct scales covering ≥ ½ of the grid, all points plotted correctly and a line of best fit or curve through the origin*

(e) **(i)** This part of the question was poorly done. Candidates were asked to determine the load when the extension is 50 mm using their graph. Some candidates did not have 50 mm on their graphs so they lost both marks. Others had difficulty reading their scales to find the position of 50 mm they also lost the marks. Some candidates earned only one mark because they did not extrapolate or make any form of making to show how they got the reading.

***Expected response**: working shown on graph and correct value from graph*

(ii) This part of the question was poorly done. Candidates were asked to calculate the spring constant, k, using the value in **(e)(i)** and the formula W = ke where W is the load, k is the spring constant and e is the extension. A majority of candidates did not convert the extension of 50 mm to 0.05 m. Other candidates failed to transpose the formula making k the subject and earned only one mark for correct substitution.

Expected response*: correct substitution of 0.05 m and their response from (**e)(i)** into the give formula and correct value.*

(f) This part of the question was fairly well done. Candidates were asked to state the relationship between the load and the extension of the spring. Some candidates wrote" Hook's Law" as their response which was not acceptable.

Expected response*: when the load(force) increases the extension also increases or the load/force is directly proportional to the extension*

(g) This part of the question was poorly done. Candidates were asked to state one source of error and the precaution taken to correct it when carrying out the experiment. Candidates gave any source of error they learnt about not necessarily an error that arise when carrying out the practical. Some common wrong responses include "place it on a level ground, read at bottom of meniscus and read correctly".

Expected response*: error of parallax - eye level / perpendicular with the reading on the rule or*
use of dirty masses - clean masses before use or
spring not close to the metre rule - use pointer/ horizontal 30 cm ruler

(h) This part of the question was not accessible to a majority of the candidates. Candidates were asked to suggest how the method used in **(b)** and **(c)** can be changed to investigate the limit of proportionality. A number of candidates did not attempt this question those that did had no idea how to plan an investigation. A small number of candidates earned only one mark for correctly describing how to tell that the limit of proportionality has been reached.

Expected response*: continue adding 50 g masses one at a time and record the extension until the spring does not return to its original length. Then plot a graph of the data and mark the point at which the direct proportionality relationship ends (limit of proportionality)*

Paper 6888/04

Alternative to Practical

General Comments

The Alternative to Practical Paper assesses the experimental skills of candidates. It consists of two questions: one Chemistry and one Physics question. The paper is marked out of 40 marks. It accounts for a 20% weighting to the overall mark of the candidates in accordance with the provisions of the syllabus. The number of entries was much higher this year relative to the previous year. The time allocated for the paper seemed to be adequate as most of the candidates attempted all the questions.

The Alternative to Practical Paper is premised on the assumption that candidates conducted experiments during the instructional phase of learning. That is why this paper assessed a range of skills including making observations; accurately reading instruments; analysing results; the use of the correct scientific technical language when making inferences, descriptions and or explanations; graphical skills; the correct presentation of significant figures when handling numerical data; to name but a few.

Performance varied across centres, but overall, the paper provided a fair test of practical skills and scientific reasoning. The highest attained score was 36 out of 40 which was higher than the previous year, with most candidates demonstrating familiarity with experimental procedures.

Comments to specific questions

Question 1

(a) **(i)** This question tested candidates' ability to read thermometer values accurately. This question was well done with some candidates making significant errors. A few candidates seemed to confuse the intervals and came up with 20.7 °C as their value and lost the mark. Common wrong responses: miscounting divisions, parallax errors.
Expected response: *27 °C*

(ii) This question required candidates to record observations of reactions between different metals and hydrochloric acid. This question was fairly done. Many responses were vague, failing to compare bubble intensity. Common wrong responses: “bubbles formed” without comparison, unclear gas test descriptions. Other candidates gave partially correct responses: "fewer bubbles" (**B**), "no bubbles" (**C**), "splint conducted, more bubbles" (**D**), with the gas test description being vague.

Expected responses: ***B*** *(zinc): moderate bubbles, slow reaction*
C *(copper): no bubbles, no reaction*
D *(magnesium): many bubbles, fast reaction*
hydrogen burns with a pop.

(iii) This question tested candidates' ability to identify the gas produced in the reaction between metals and hydrochloric acid. Most candidates correctly identified hydrogen, demonstrating good understanding of gas tests. However, some candidates incorrectly identified the gas as oxygen, argon and carbon dioxide showing confusion about gas identification.

Expected response: *hydrogen / H_2*

(iv) This question required candidates to identify the most and least reactive metals based on their observations. This question was generally well answered with most candidates correctly identifying the most and least reactive metals. There were few candidates who gave iron and zinc as the most or least reactive metal.

Expected response: *most reactive - magnesium*
least reactive - copper

(v) This question required candidates to state two reasons justifying their identification of the most and least reactive metal in **(a) (iv)**. Most candidates were able to get the two marks. Some candidates gave responses like magnesium is more reactive because it is higher in the reactivity series instead of using the observations in **(a) (ii)**. Others gave the first reason as magnesium produced more bubbles and the second reason as copper produced no bubbles which made them to score one mark. The other reason was supposed to be of temperature change either for magnesium or copper. Some gave unclear statements like “it reacted more” without specifying how. Some candidates were confusing reactivity with solubility, with some candidates mentioning that it dissolved faster.

Expected response: *highest temperature change for magnesium or lowest temperature change for copper.*
more bubbles/ fizz/ fastest reaction for magnesium or less bubbles/ fizz/ slowest reaction for copper.

(vi) In this question candidates were asked to identify a variable that was controlled to ensure the experiment was a fair test. This question was fairly done. Most candidates were giving temperature not specifying whether it’s the initial of final temperature and they lose the mark.

Some suggested that the volume of the metals must be the same or used the same type of metal which was the independent variable.

Expected response*: same mass of metals / same concentration of acid / same initial temperature of acid.*

(b) (i) The question required the candidates to suggest and record two observations the student made during the reaction of aqueous silver nitrate with the metals iron, copper and magnesium This question was poorly done. Most candidates failed to see that the question needs two observations. They will give a grey precipitate and leave the rate at which the precipitate will be formed, resulting in the candidates losing all the marks. Some candidates gave colours of the metal compounds as they appear in the chemistry practical notes, for example copper(II) compounds are blue, so the candidates will write a blue precipitate insoluble in excess. Other common wrong responses were white precipitate, formation of bubbles or gas.

Expected response: *iron: grey precipitate formed slowly*
copper: grey precipitate/ solution formed very slowly
magnesium: grey precipitate formed quickly.

(ii) The candidates asked to state and explain the type of reaction that occurs using the results in Table 1.2. This question was fairly done with most candidates stating that the type of reaction is exothermic and failing to explain why. Most candidates would use theory as their explanation for example "more heat is released than being absorbed" or in reference to bond breaking and formation. Some candidates assumed that the reaction is endothermic, explaining that the rise in temperature means that heat is being absorbed from the environment that is why the temperature is increasing. Other common wrong responses include chemical reaction, physical change, combination reaction, neutralisation reaction.

Expected response: *exothermic reaction because heat is released causing an increase in temperature.*

(iii) The candidates were told that zinc nitrate is one of the substances formed when zinc granules react with aqueous silver nitrate. They were asked to name the other substance formed in the reaction. This question was fairly done. Some candidates incorrectly wrote hydrogen or oxygen gas confusing the displacement reaction with reactions involving acids and water. Other common incorrect responses include: zinc oxide, silver ion, silver nitrate, zinc nitrate and zinc.

Expected response: *silver*

(iv) This question required the candidate to state and explain which is the least reactive metal using the results in Table 1.2. This question was fairly done with most candidates getting the first mark correctly for stating the least reactive metal but lost the second mark since they were not using the information in Table 1.2 but the reactivity series to explain why copper is least reactive. Candidates were also explaining using the rate of the formation of bubbles in Table 1.1

Expected response: *copper because the grey solid formed slowest or had lowest maximum temperature.*

(c) Candidate were asked to arrange the four metals in order of reactivity starting with the most reactive metal. This question was well done, candidates were able to arrange the metal in order of reactivity. Some candidates gave zinc as the least reactive. In Table 1.2 it is given that zinc formed grey precipitate slowly, so candidates assumed that it means zinc is less reactive. Some candidates would simply arrange the metals in alphabetical order (copper, iron, magnesium, zinc) instead of using the experimental data.

Expected response*: magnesium, zinc, iron, copper*

Question 2

(a) The candidates were given a set-up of experiment where a student was investigating the relationship between a load and the extension of a spring. This question required the candidate to read and record the position of the bottom of the spring, X_0 , and write the answer in millimetres. Generally, well done as a majority of the candidates were able to correctly read the scale and convert the 55 cm to 550 mm and earn the mark. However, some candidates incorrectly identified the reading as 55,000 mm, 55 mm, 5.5 mm showing significant unit conversion error. Others incorrectly identified 54 mm, suggesting a misreading of the scale. Common errors included unit conversion mistakes and misreading of measurement scales.

Expected response*: 550 mm*

(c) This question required candidates to record the new position of the spring after adding a 100 gram mass. Generally, well done as quite a number of candidates were able to correctly read the value and convert it to millimetres as 577 mm on the workspace on the question paper but failed to follow instruction to transfer it to Table 2.1. They ended up looking at the trend of the new position 'X' in Table 2.1 that the figures differ by 20 mm and hence wrote 575 mm which was wrong. However, some candidates who had a challenge of converting cm to mm will give responses like 57.7 mm, 57.0 mm etc which were wrong. Some candidates subtracted from the baseline, leading to negative or unrealistic extensions.

Expected response: *577 mm*

(c) The student repeats the procedure **(b)** by adding 50 g masses, one at a time, on the suspended mass. They records the new positions of the bottom of the spring 'X' in Table 2.1.

(i) Candidates were asked to complete Table 2.1 by filling in the unit for the extension. The question was fairly done as some candidates were able to state the correct unit of extension as mm although some candidates were confused and wrote units like cm, m, N/m, e which did not score a mark. Some candidates left blank spaces even though they have written the units on previous questions.

Expected response*: cm or mm*

(ii) Generally, well attempted as the majority of the candidates were able to calculate the load from the given mass on the table using the given formula to get 1.0 N, 1.5 N, 2.0 N, 2.5 N and 3.0 N but some candidates failed to transfer the values to Table 2.1 and left the values on the workspace on the question paper. Some candidates got these correct values and further multiplied them by ten to get 10 N, 15 N, 20 N, 25 N and 30 N which did not earn any mark.

Expected response*: 1.0 N, 1.5 N, 2.0 N, 2.5 N, 3.0 N*

(iii) The candidates were asked to calculate the load, W, for each mass using the formula W= mg. The question was fairly done as some candidates were able to subtract the answer in (a) i.e. 550mm from the values of the new position X to get 27 mm, 45 mm, 65 mm, 85 mm and 105 mm provided they have recorded the first value of 577 mm on Table 2.1 obtained full marks. Some candidates who had a challenge with **Q2(a)** ended up subtracting their wrong, X_0, from the new position X and came up with wrong values. Some ended up subtracting the consecutive values of the new position 'X' and end up with a constant value of 20 mm for all the different loads which did not score.

Expected response*: 27, 45, 65, 85, 105*

(d) Candidates were asked to plot a graph of extension against load using the results in Table 2.1. More candidates were able to get at least one mark from this question as they were able to plot points using data from the table. Most candidates could not to come up with the correct scale on both axes which resulted in loss of one mark for the wrong scale. Some used the points from their table when plotting which resulted in a nonlinear scale and they lost most of the marks. A few candidates would plot points that were not in table 1.2 and lost most or all of the marks. Those who had negative values in their data struggled to make their graph.

Expected response: *correct scales covering ≥ ½ of the grid, all points plotted correctly and a line of best fit or curve through the origin*

(e) **(i)** The candidates were asked to use the graph to determine the load when the extension is 50 mm. This part was fairly done by the majority of the candidates that were able to correctly plot and join their points from Table 2.1 hence obtaining the maximum marks if they correctly read the value of the corresponding load. Some candidates that did not draw the line of best fit as this skill needed the graph to be present to earn the marks. A few candidates used non-linear scales or values that are non-existent from Table 2.1 also could not score. A reasonable load value was approximately 1.6 N obtained from correct data used in **Q2(d)**. Some candidates failed to correctly read the load from their scales hence coming up with values such as 1.52 N, 2.5 N etc which did not earn credit.

Expected response: *working shown on graph and correct value from graph*

(ii) Candidates were asked to calculate the spring constant, k, using the values in **e(ii)** and the extension of 50 mm. This question was fairly done as most candidates were able to get one mark from this question. A majority of the candidates failed to earn the first mark since they were not able to convert 50 mm to metres i.e. 0.05 m to get the spring constant. Some candidates divided by 0.5 m or multiplied by 50 mm thus lost the marks.

Expected response: *correct substitution of 0.05 m and their response from (**e)(i)** into the give formula and correct value*

(f) The candidates were asked to state the relationship between the load and the extension of the spring. Most candidates did well on this question showing a good understanding between the relationship between extension and load that they are directly proportional which the fundamental principle of Hooke's Law is. Some candidates correctly described "As load increases, extension increases," which was conceptually correct though less precise. However, some candidates failed to express themselves correctly as they will say "high load results in high extension" not showing the concept of change in the quantities thus they could not score.

Expected response*: directly proportional/ increase in force, increase in extension*

(g) **(i)** This question required candidates to identify a possible source of error in the experiment. The question was fairly done as some candidates correctly mentioned the error of parallax though some used a wrong spelling parallel error which did not score. Surprisingly none of the candidates mentioned the use of dirty masses or spring not close to the metre ruler as sources of errors when carrying out this experiment. Some candidates gave incorrect responses including reading the scale incorrectly, ruler not starting from zero i.e. zero error, use of a broken ruler to mention a few which did not earn a mark. Some mentioned precautions that are taken when reading the volume of a liquid such that the reading should be taken at the bottom of the meniscus.

Expected response: *error of parallax / dirty masses / spring not close to the ruler*

(ii) This question tested candidates' ability to suggest appropriate precautions to minimise errors. The question was quite challenging to the candidates as they had to address their correct error in **g(i)** i.e. eye level with the reading on the ruler to avoid parallax error, use clean masses to avoid added mass, use a pointer or horizontal 30cm ruler to point to the exact value. Even those that got the parallax error correct in **(g)(i)** got this one wrong as they will say that "the eye must be parallel to the reading or perpendicular to the ruler or scale which is wrong. Common wrong responses include "repeat the experiment, make sure the spring is not moving before taking the reading, make sure the spring has not reached the limit of proportionality, the experiment should be done on a flat surface, use a ruler that is not broken".

Expected response: *error of parallax - eye level / perpendicular with the reading on the rule or*
use of dirty masses - clean masses before use or
spring not close to the metre rule - use pointer/ horizontal 30 cm ruler

(h) Candidates were asked to suggest how the method described in **(b)** and **(c)** can be changed to investigate the limit of proportionality. Generally, this question was poorly done as majority of candidates either managed to score one mark or zero as they failed to describe how to find the limit of proportionality. Common wrong responses were that add the 50g masses until the spring stops stretching which is wrong since that means the spring has already gone beyond the limit of proportionality. Some thought that using bigger loads or adding the masses at once will show the limit of proportionality without describing how. Other candidates incorrectly suggested "stretch spring by hand until it cannot return," demonstrating misunderstanding of the concept. Others gave a partially correct response "suspend more masses until spring changes," which showed some understanding but lacked precision.

Expected response: *continue adding 50 g masses one at a time and record the extension until the spring does not return to its original length. Then plot a graph of the data and mark the point at which the direct proportionality relationship ends (limit of proportionality)*